

MOST REPEATED QUESTIONS

CBSE Class 12 Chemistry • Analysis of 3 Paper Sets

By Pragya

1. Physical Chemistry

Calculate EMF and ΔG using the Nernst Equation

Question: Calculate the emf and/or ΔG° for a given cell reaction (e.g., $\text{Zn}|\text{Zn}^{2+}||\text{H}^+|\text{H}_2$) at a specific temperature.

Frequency: 3 out of 3 papers (Repeats in every set)

Conductivity and Molar Conductivity (Kohlrausch Law)

Question: Calculate the molar conductivity (Λ_m) and degree of dissociation (α) given the conductivity (κ) and concentration. Often involves explaining the effect of dilution on conductivity.

Frequency: 3 out of 3 papers

Colligative Properties (Boiling/Freezing Points)

Question: A numerical problem requiring the calculation of the mass of solute, molar mass, or the change in freezing/boiling point. These questions frequently involve the **Van't Hoff factor (i)**.

Frequency: 3 out of 3 papers

First Order Kinetics Proof

Question: Show that for a first-order reaction, the time required for **99% completion is twice** the time required for **90% completion**.

Frequency: 2 out of 3 papers

2. Organic Chemistry

Cannizzaro Reaction

Question: Write the reaction or identify products for an aldehyde with no α -hydrogen (like Benzaldehyde or Formaldehyde) when treated with concentrated alkali.

Frequency: 3 out of 3 papers

SN1 vs SN2 Reactivity

Question: Arrange a given set of haloalkanes in increasing order of reactivity towards SN1 or SN2 mechanisms, or account for why a specific compound is more reactive (often involving steric hindrance or solvent polarity).

Frequency: 3 out of 3 papers

Distinction Tests (Aldehydes, Ketones, Amines)

Question: Give a chemical test to distinguish between pairs of compounds.

- **Iodoform Test:** Distinguishes methyl ketones ($\text{CH}_3\text{-C=O}$) or methyl carbinols.
- **Tollens' Test:** Distinguishes aldehydes from ketones.

Frequency: 3 out of 3 papers

Sandmeyer's Reaction

Question: Write the chemical equation for Sandmeyer's reaction (conversion of diazonium salts to haloarenes).

Frequency: 2 out of 3 papers

3. Inorganic Chemistry

Trends in d-Block Elements

Question: Account for specific trends such as:

- Why Mn^{3+} is a strong oxidizing agent or $E^\circ (\text{Mn}^{2+}/\text{Mn})$ is highly negative.
- Variation in melting points of transition metals.

Frequency: 3 out of 3 papers

Coordination Compounds (CFT & Color)

Question: Explain the color of a complex (like $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$) using **Crystal Field Theory (CFT)**.

Frequency: 2 out of 3 papers

Lanthanoid Contraction

Question: Explain the cause and consequences of lanthanoid contraction (e.g., similar atomic radii of Zr and Hf).

Frequency: 2 out of 3 papers

4. Biomolecules

Reactions of Glucose

Question: Write the reaction of D-Glucose with **HCN** (confirming carbonyl group) or other reagents like Bromine water.

Frequency: 2 out of 3 papers